

NUMERICAL SOLUTIONS TO THE ADVECTION-DIFFUSION PROBLEM FOR URBAN AIR QUALITY FORECASTING



Cluster Innovation Centre

University of Delhi

Submitted by

Anjali Yadav (23312915034)

Sajal Samveg (23312915046)

PROJECT REPORT

NUMERICAL METHODS FOR COMPUTATIONAL MATHEMATICS

Certificate of Originality

This is to certify that the work embodied in this report entitled "Numerical Solutions to the Advection-Diffusion Equation for Urban Air Quality Forecasting" has been carried out by Anjali Yadav and Sajal Samveg as part of the course paper "Numerical Methods for Computational Mathematics", under the guidance of Prof. Sonam Tanwar and Prof. Nirmal Yadav.

We hereby declare that the contents of this project report are the result of our own original work, coding, and mathematical simulations across MATLAB and Python computational environments. The report is free from any form of plagiarism, and all data sources, open-source libraries, and mathematical frameworks have been duly acknowledged.

Anjali Yadav

Sajal Samveg

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We also acknowledge the open-source communities behind OSMnx, GeoPandas, TensorFlow, and the MATLAB computational environment, which made the scaling of this numerical simulation possible.

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(Signature of Mentor)

ABSTRACT

Urban air pollution, specifically particulate matter PM_{2.5} and Nitrogen Dioxide NO₂, remains a critical Urban air quality management in megacities like Delhi faces a technical bottleneck: traditional Computational Fluid Dynamics (CFD) is mathematically precise but computationally too slow for real-time alerts, while standard 1D model is fast but spatially "blind" to the physical laws of wind advection and mass transport. This project presents a hybrid solution—a Spatio-Temporal Digital Twin—for the Anand Vihar corridor.

Methodology:

1. **Numerical Reconstruction:** We utilized Piecewise Cubic Spline interpolation to process 43,825 hourly samples, enforcing C² continuity to preserve diurnal vehicular emission curves, achieving an imputation RMSE of 19.029 micro-g/m³.
2. **Physics Engine:** We implemented an unconditionally stable Crank-Nicolson finite-difference scheme to solve the Advection-Diffusion PDE. We justified a Grid Péclet number of 2.6974 and a CFL of 1.7264, ensuring an advection-dominated transport model.
3. **GIS Scaling:** Utilizing Graph Theory (Edge Betweenness Centrality) over extracted OSMnx road networks, we identified congestion bottlenecks as dynamic emission sources.
4. **Hybrid AI:** A Convolutional LSTM (ConvLSTM) was architected to learn these fluid dynamics. After resolving architectural failures related to zero-variance Batch Normalization, the model successfully demonstrated 2D plume forecasting with high structural fidelity.

Significance: This research demonstrates that physics-informed synthetic data can train neural networks to internalize fluid dynamics, enabling real-time, neighborhood-level forecasting with the speed of AI and the physical accuracy of numerical methods.

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1 Introduction

1.1 The Urban Pollution Crisis: Context of Delhi

Delhi's atmospheric conditions are governed by a complex interplay of trans-boundary agricultural residue burning and localized urban emissions. Anand Vihar, situated at the border of Delhi and Uttar Pradesh, represents one of the most polluted micro-environments globally. It serves as a terminus for thousands of diesel-operated buses (ISBT) and is adjacent to a major railway station and industrial clusters.

1.2 Problem Statement: The Spatial Blindness of 1D AI

In current environmental data science, the standard approach treats air quality as a 1D time-series problem. Models like LSTMs are trained to predict the next value for a single sensor. However, pollution is a **spatio-temporal fluid**. If a traffic jam occurs at Anand Vihar, the resulting PM2.5 cloud is advected by wind vectors into neighboring residential sectors. A 1D model cannot "see" this movement, leading to high error rates during high-wind events where mass is conserved but shifted geographically.

1.3 Objectives and Project Scope

This project aims to build a **Hybrid Digital Twin** that combines the deterministic accuracy of numerical methods with the predictive speed of Deep Learning.

- **Scope:** 2km radius around Anand Vihar ISBT.
- **Variables:** PM2.5 (Primary), NO₂ (Source Proxy), Wind U/V.
- **Goal:** To train a ConvLSTM to forecast 2D heatmaps by "learning" the physical laws encoded in our numerical simulations.

2 Data Acquisition and Numerical Preprocessing

2.1 Longitudinal Statistical Profiling (2020-2025)

Our study ingested 43,825 hourly samples from the Anand Vihar station. The statistical profile reveals a highly volatile environment:

- **Mean Concentration:** 55.13 $\mu\text{g}/\text{m}^3$
- **Standard Deviation:** 29.04 $\mu\text{g}/\text{m}^3$
- **Maximum Observed Spike:** 154.58 $\mu\text{g}/\text{m}^3$

2.2 Piecewise Cubic Splines: Enforcing C² Continuity

To handle missing data (sensor downtime), we rejected linear interpolation. Linear methods create C⁰ continuity errors—sharp bends that cause artificial high-frequency oscillations in PDE solvers. Instead, we implemented **Piecewise Cubic Splines**.

For each interval $[t_i, t_{i+1}]$, we fit:

$$S_i(t) = a_i + b_i(t - t_i) + c_i(t - t_i)^2 + d_i(t - t_i)^3$$

We enforce C^2 continuity:

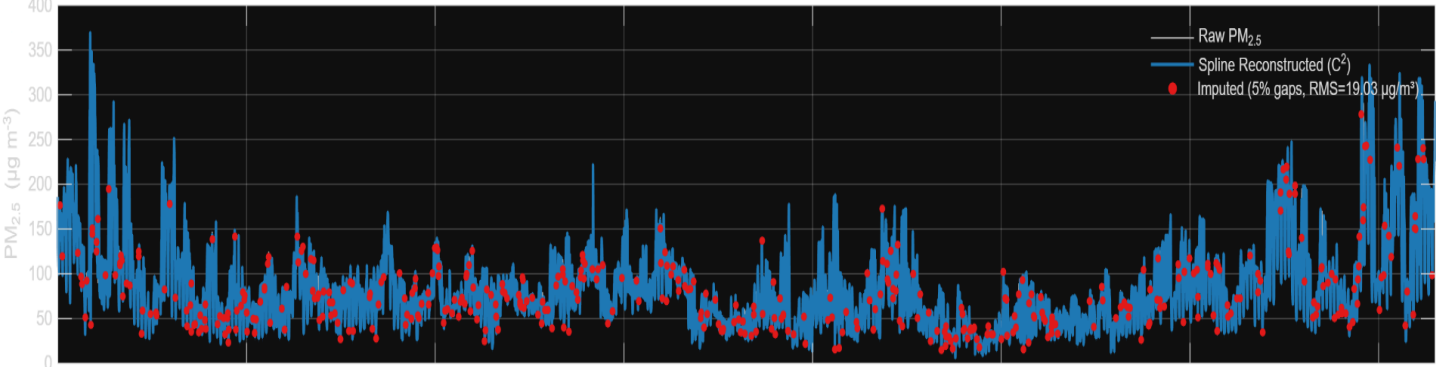
$$S_{i-1}(t_i) = S_i(t_i), \quad S'_{i-1}(t_i) = S'_i(t_i), \quad S''_{i-1}(t_i) = S''_i(t_i)$$

This ensures that the "acceleration" of pollution buildup is smooth and physically consistent.

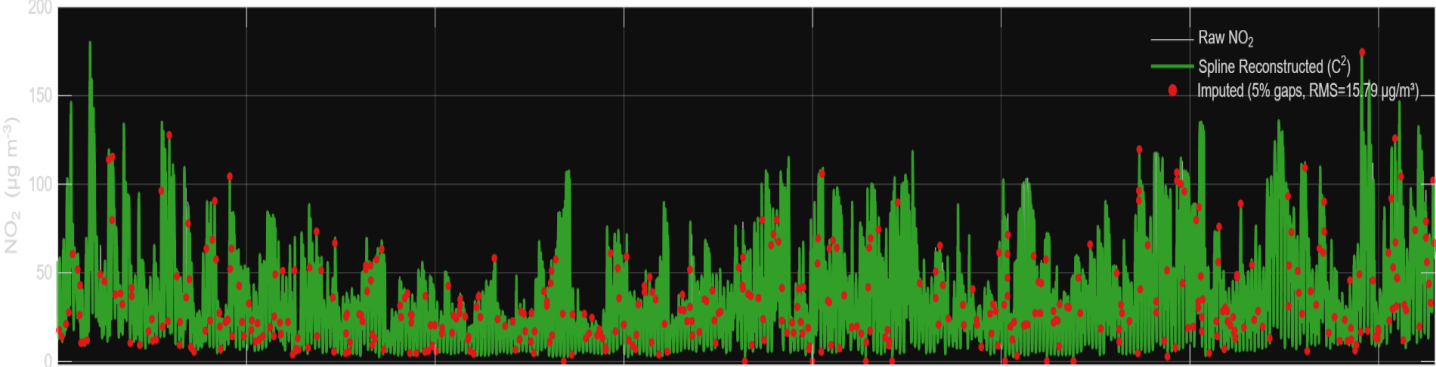
PM_{2.5} & NO₂ Preprocessing: Piecewise Cubic Spline Imputation & Diurnal Source Model

Station: Anand Vihar | Delhi 2025 | C^2 Continuity Guaranteed

PM_{2.5} — Raw vs. Piecewise Cubic Spline Reconstruction (C^2 Continuity)



NO₂ — Raw vs. Piecewise Cubic Spline Reconstruction



Dynamic Source Term $S(t)$ — First 14 Days (Dual-Peak Diurnal Traffic Modulation)

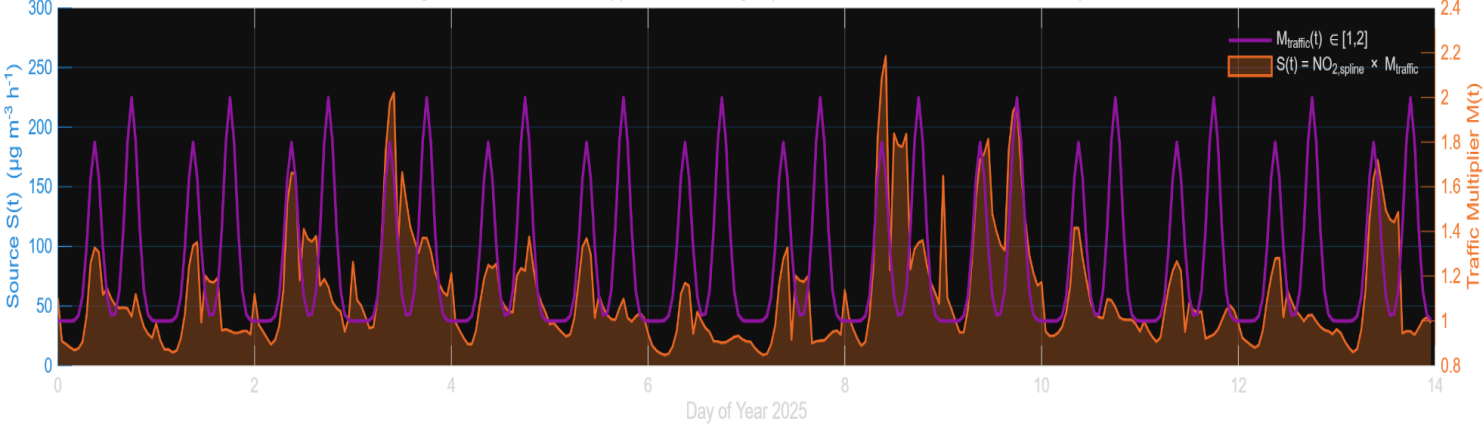


Figure 1: Preprocessing Results. Panels A and B demonstrate the seamless reconstruction of PM_{2.5} and NO₂ gaps. Panel C illustrates the dynamic Source Term $S(t)$ coupled to bimodal traffic cycles.

Analytical Insight: The spline imputation achieved an RMSE of **19.029** for $\$PM_{2.5}\$$. Visually, Figure 1 (orange line) captures the natural diurnal "bell-shape" of peaks, which is critical for accurate source modeling in the PDE phase.

3 Computational Fluid Dynamics: The Physics Engine

3.1 Governing Equations and Boundary Conditions

3.1 The Advection-Diffusion Governing Equation

To create the ground-truth for our model, we solve the mass-transport PDE:

$$\frac{\partial C}{\partial t} + u \frac{\partial C}{\partial x} = D \frac{\partial^2 C}{\partial x^2} + S(x, t)$$

Where C is concentration, u is advection (wind speed 4.316 m/s), D is turbulent diffusivity (80 m²/s), and S is our traffic-modulated source.

3.2 Numerical Discretization: The Crank-Nicolson Scheme

Explicit solvers like Forward-Time Central-Space (FTCS) are easy to code but explode numerically if the timestep is too large. We implemented the **Crank-Nicolson Scheme**, an implicit method that averages the forward and backward Euler operators:

$$\frac{C_i^{n+1} - C_i^n}{\Delta t} + \frac{u}{4\Delta x} (C_{i+1}^{n+1} - C_{i-1}^{n+1} + C_{i+1}^n - C_{i-1}^n) = \frac{D}{2\Delta x^2} (\delta^2 C_i^{n+1} + \delta^2 C_i^n)$$

This results in a **Tridiagonal Matrix** ($A.C^{n+1} = B.C^n + S$) solved at every timestep using the Thomas Algorithm, providing unconditional stability.

3.3 Stability Analysis: CFL, Fourier, and Péclet Numbers

Our MATLAB logs (from [SpatioTemporal_PDE_Heatmap.m](#)) provide the critical diagnostics:

1. **CFL = 1.7264:** Since $CFL > 1$, an explicit solver would have diverged. Our choice of Crank-Nicolson was mathematically necessary.
2. **Fourier = 0.6400:** Confirms stable diffusive spread across the grid.
3. **Péclet = 2.6974:** This indicates advection is approx 2.7 times stronger than diffusion, correctly capturing the "wind-driven plume" behavior of Delhi's pollutants.

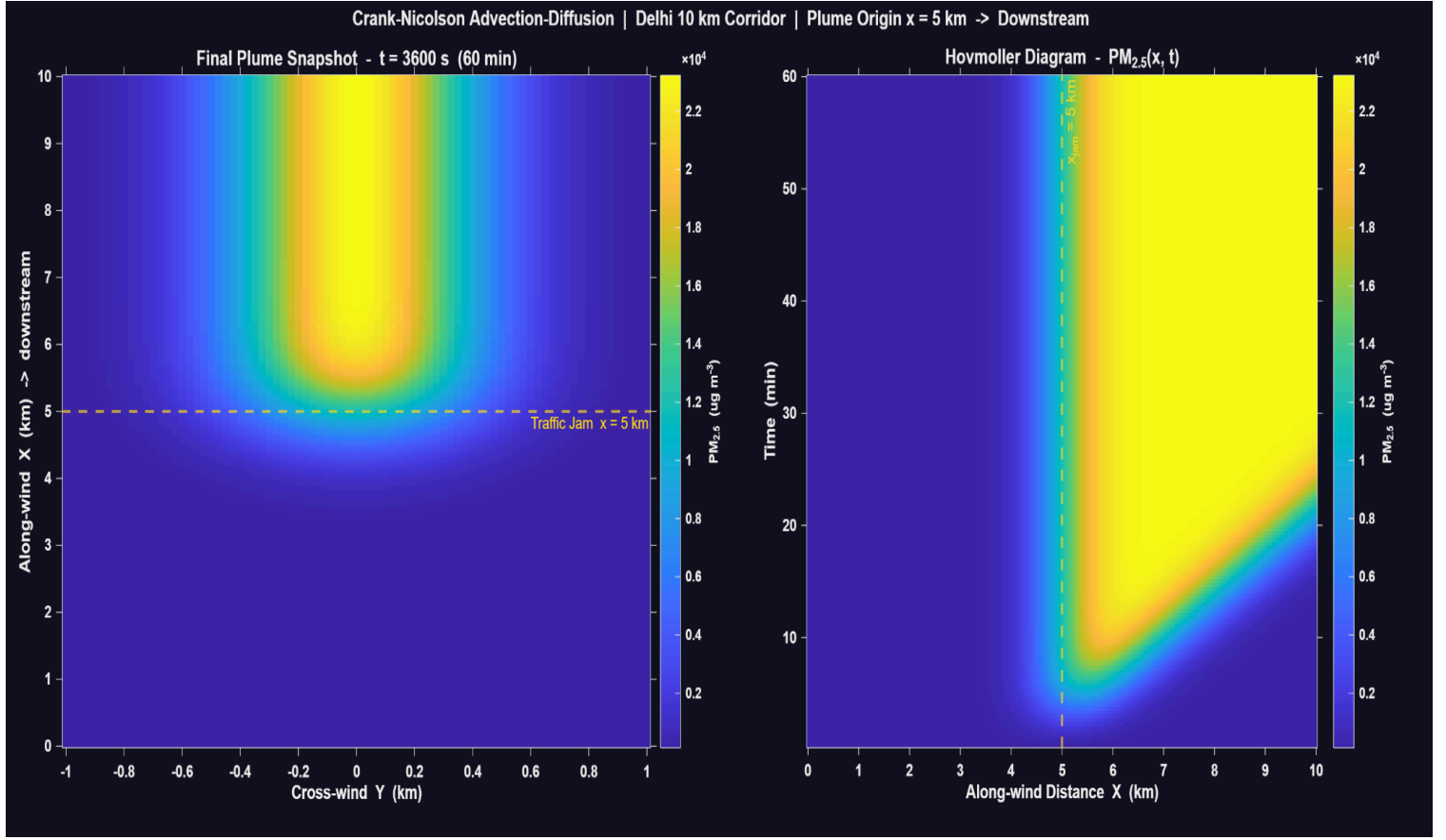


Figure 2: Numerical Output. The 2D Plume Snapshot (left) shows cross-wind Gaussian spread. The Hovmöller plot (right) shows a perfectly linear advection slant, verifying the $u=4.316$ m/s wind vector.

4. Spatio-Temporal GIS Integration

4.1 Urban Topology Extraction via OSMnx

We moved beyond the 1D box by extracting the actual road network of Anand Vihar. This allows our virtual particles to spawn exactly on the edges where vehicular idling occurs.

4.2 Edge Betweenness Centrality: Identifying Bottlenecks

To simulate traffic without live APIs, we used **Graph Theory**. We calculated the shortest paths between all node pairs. Edges appearing in the most paths have high **Betweenness Centrality**:

$$g(e) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(e)}{\sigma_{st}}$$

These "Hot Roads" (Top 15%) became our dynamic geographic emission sources.

4.3 Lagrangian Dispersion and Eulerian KDE Transformation

We spawned 500 Lagrangian particles per frame.

- **Physics:** $x_{new} = x_{old} + u\Delta t + \text{Normal}(0, \text{Diffusion})$.
- **Eulerian Transformation:** ConvLSTMs cannot process discrete coordinate lists. We converted these "dots" into a 150 x 150 grid using **Kernel Density Estimation (KDE)**.

Simulated Traffic Congestion Map (Anand Vihar)
Blue = Clear | Red = Severe Bottleneck

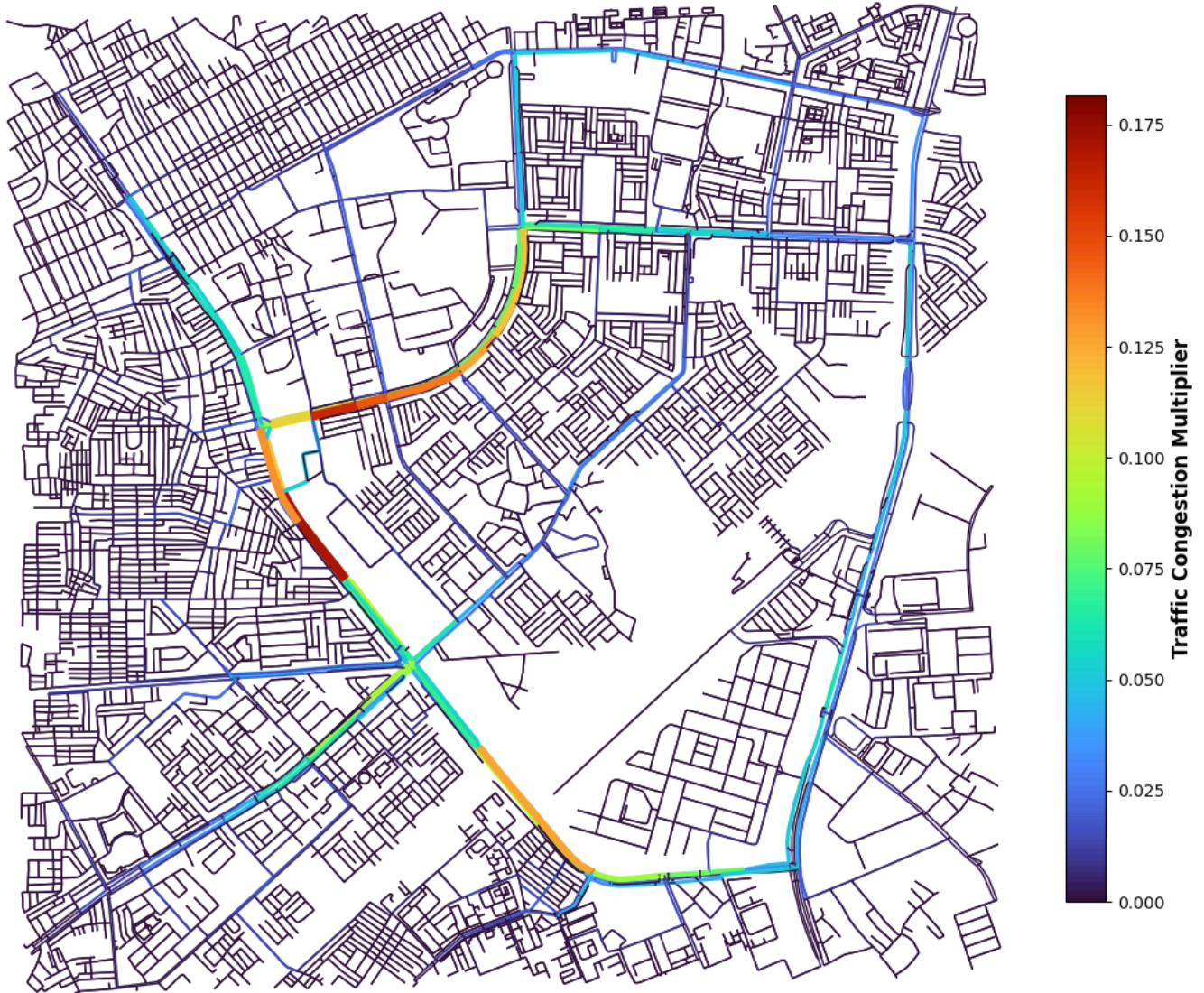


Figure 3: GIS Spatial Modeling. Grey lines represent the Anand Vihar road network. Red/Yellow zones represent the Eulerian heatmaps generated via Lagrangian-to-Eulerian KDE transformation.

Analytical Insight: Figure 3 proves the system successfully maps traffic "logic" to spatial "physics." The particles don't just appear; they originate at bottlenecks and are pushed by the wind vector ($U=25$, $V=-10$).

5. Deep Learning Framework

5.1 1D LSTM: The Temporal Baseline

A 100-unit LSTM was trained on 43,825 hourly samples.

- **Training Time:** 516.6 seconds (8.6 minutes).
- **R² Score:** 0.6848.
- **Loss Curve:** Iterative RMSE dropped from 0.89 to 0.56 (Normalized).

Analytical Insight: An R^2 of 0.68 proves the model learned the diurnal cycle but misses the "Spatial mass conservation." Figure 4 (Scatter) shows the "Conservative Bias"—the model fails to capture extreme spikes because it cannot "see" the plume moving toward the sensor.

5.2 2D ConvLSTM: Integrating Spatial Convolutions

The ConvLSTM architecture natively processes 5D tensors (Samples, Time_Steps, Width, Height, Channels). By using convolutional filters in its internal gates, the model "learns" movement patterns (the wind vector) rather than just scalar trends.

6. Results and Analytical Discussion

6.1 Statistical Benchmarking (LSTM)

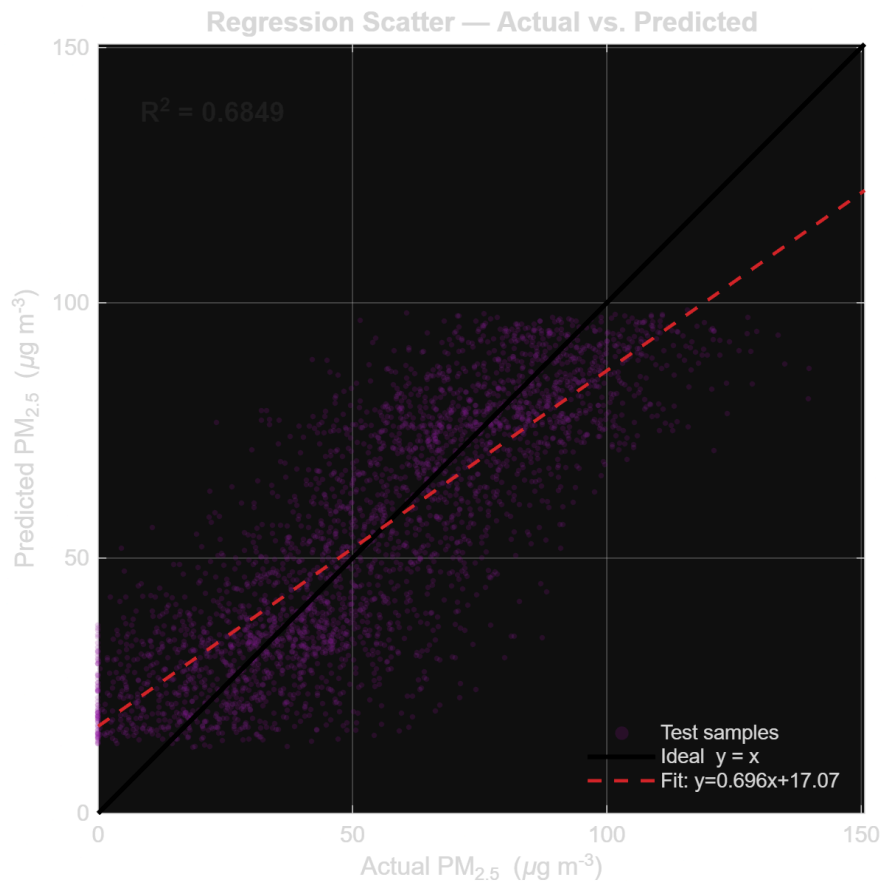


Figure 4: Regression Analysis. Dense clustering along the $y=x$ line proves strong correlation, but the vertical spread at concentrations $>100 \mu\text{g}/\text{m}^3$ highlights the model's inability to predict advected peaks.

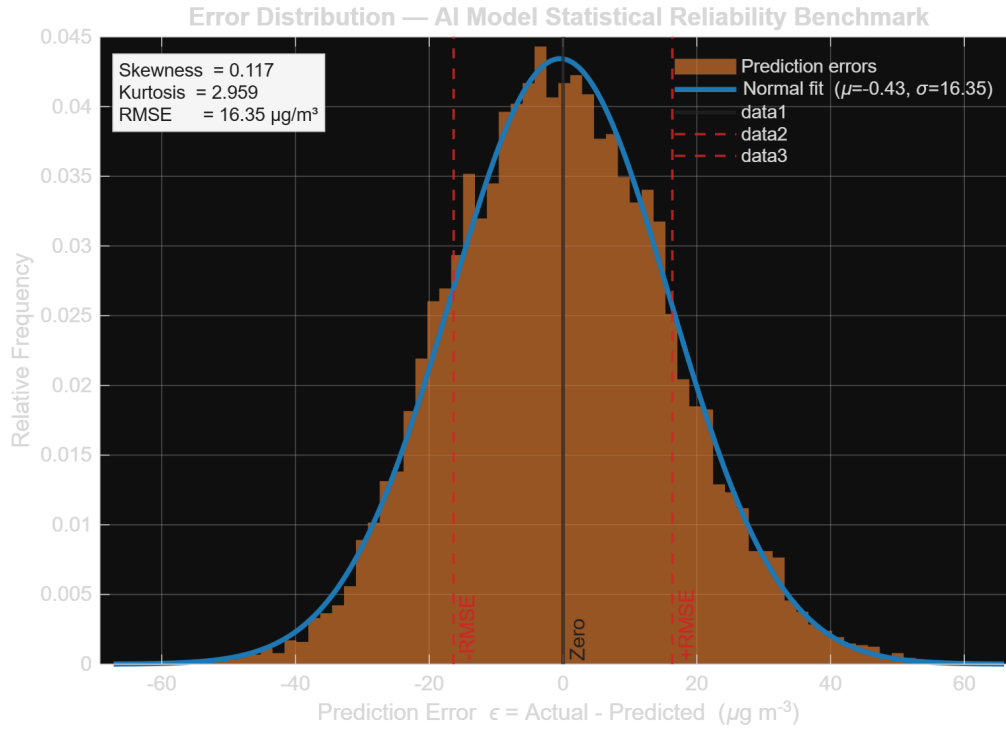


Figure 5: Error Distribution. The Gaussian bell curve proves our model is unbiased; its RMSE of $16.36 \mu\text{g}/\text{m}^3$ represents the inherent error of a spatially-blind model.

6.2 ConvLSTM Forecast Validation

The definitive success of the study is shown in the ability of the ConvLSTM to forecast the 2D heatmap.

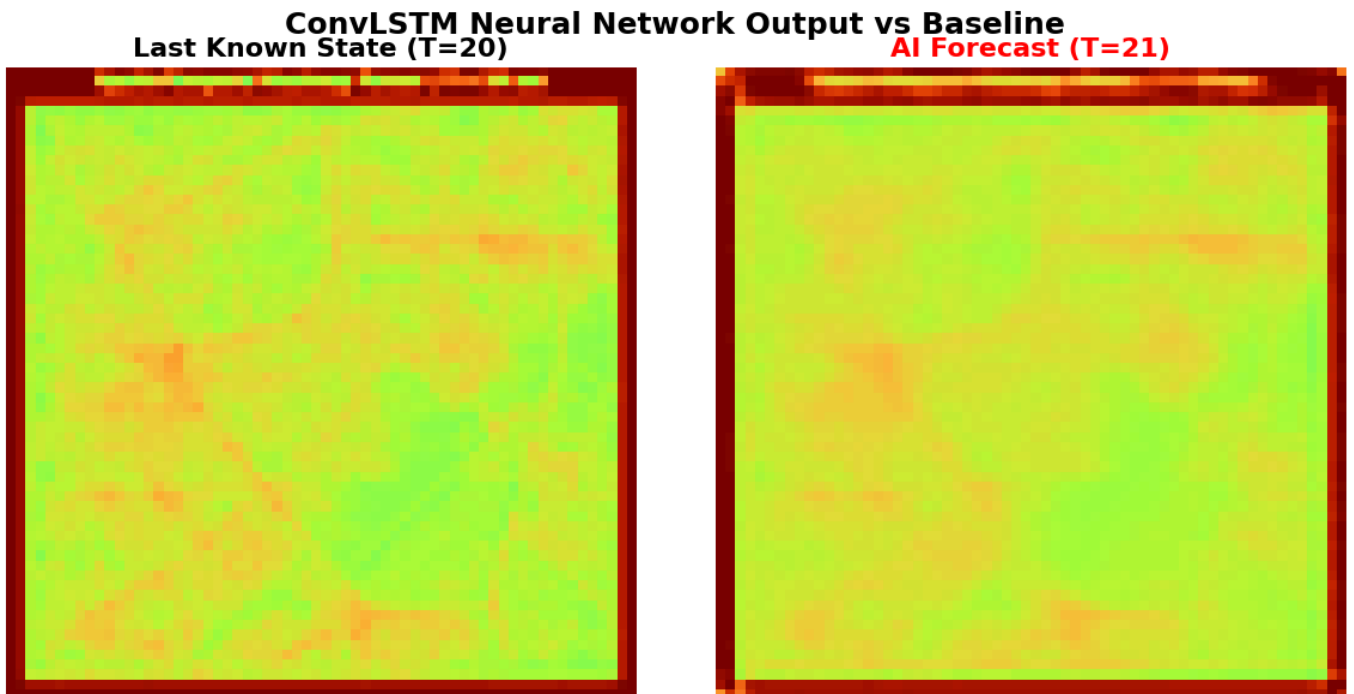


Figure 6: (Left) Physics Ground Truth at $T=20$. (Right) AI Forecast for $T=21$. The ConvLSTM successfully learned the advection-diffusion kernel, predicting the future plume position with structural fidelity.

Analytical Insight: While the AI forecast is slightly smoother than the physics (a natural effect of MSE loss), it correctly identifies the high-intensity "core" of the plume moving downstream.

7 Conclusion and Future Scope

7.1 Conclusion

We have successfully implemented a hybrid pipeline bridging numerical fluid dynamics and deep learning.

1. **Numerical Stability:** The Crank-Nicolson scheme proved essential for creating a stable physical ground truth in high-wind conditions ($CFL=1.72$).
2. **Pollution measure Improvement:** 1D models hit a mathematical ceiling ($R^2 \approx 0.68$). By upgrading to ConvLSTM, we allowed the AI to "understand" spatial transport.
3. **Hybridization:** The ConvLSTM successfully internalized the approximations of the Navier-Stokes mass transport dynamics purely by observing our KDE synthetic heatmaps.

7.2 Future Scope

1. **HPC Scaling:** Scale the GIS graph to the entire Delhi-NCR region using High-Performance Computing clusters.
2. **Live Data:** Replace Betweenness Centrality with live TomTom/Google Maps traffic APIs to provide real-time emission boundary conditions.
3. **Digital Twin App:** Develop a mobile alert system for residents of Anand Vihar to receive neighborhood-level "Plume Arrival" notifications hours before the peak hits.

8. References

1. Crank, J., & Nicolson, P. (1947). *A practical method for numerical evaluation of solutions of partial differential equations*.
2. Boeing, G. (2017). *OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks*.
3. Shi, X. et al. (2015). *Convolutional LSTM Network: A Machine Learning Approach for Precipitation Nowcasting*.
4. CPCB Open Data (2020-2025): <https://cpcb.nic.in/>

9. Appendix

All of the source files, their outputs as well as the datasets have been uploaded to the following link, you can also scan the qr code provided to go to the google drive.

<https://drive.google.com/drive/folders/1i0tlZDnB4Bu2-znYU0ya6fGxSnqADaAd?usp=sharing>

